Preface

If you picked up this book you are probably one of those lucky people who run plants. Either a thinly spread engineer (branch of specialty is irrelevant), a newly promoted technical manager, or a harassed technologist or senior mechanic, who just was told: "See that plant out there? You're in charge of making it work!" Even if you've been in plants for years, that's enough to make your innards rumble. If you have just stepped out of school, into your first plant, or into a totally different plant from the previous one you were at, your reaction might be more severe, especially if you belong to one of the numerous organizations with no budget for rotating machinery specialists (who look after what moves a process through its paces) or environmental specialists (who make certain you don't get fined or jailed, good intentions notwithstanding, as you run your plant). At this point, I should explain where I fit in with your agenda.

Twenty-some years ago, some heated arguments on the subject of how much I wanted to be a rotating machinery specialist took place in Canada's wild and woolly north. I was fencing with my boss, a process engineer, who was recommending I join his field. It was what my company needed, he asserted. I thought it needed both of us doing what we loved best. My career bears witness to the fact that I won the match, in the short- and long-term.

Time since has brought a few things forcibly home to me. To start with, the more I dealt with plant machinery in any form, the more I accepted that process conditions could affect the performance of that machinery at least as much as actual mechanical characteristics. In operations, repair and overhaul, or retrofit design and reengineering, what keeps people like me a step ahead of the manufacturer's field service representatives is knowledge of the process and familiarity with the controls that govern the entire system. In turn, the process engineer who gets handed a plant to run must acquire some basic knowledge of my bread and butter, the machinery that makes everything move up, down, or around. In large facilities, such as the ones I was fortunate enough to spend time in, there generally are in-house rotating machinery specialists. Often, though, the process engineer is not that lucky and gets everything—process components, machinery, controls, and all.

Life handed me an education (after formal degree acquisition) in rotating machinery specialization and the environmental technology that goes with it (yes, we machinery "cranks" run the stuff that turns out arguably 80 percent of the gunk in the universe). While doing this, I worked with scores of process engineers, control engineers, and various other specialists on a variety of projects that were among the most high profile in the world in their own right. It was "arranging to be in the right place—an operating plant—to get the best education in the best school in the world."

After all, curriculum, undergraduate or otherwise, is not necessarily any comfort. In my day they rarely taught this stuff to process, chemical, or mechanical engineers

at universities. They still don't. That leaves all the young engineers in the same boat—without any practical guide for reference.

My editor at McGraw-Hill was keen that they should have one, and one that was easy to read. We soon found we were on the same page on the subject of readability. We do not like technical material that sounds more intellectual than it absolutely has to, and we do like many diagrams, photographs, tables, and figures.

I add two other ingredients to my books and courses: (1) information on items (such as condition monitoring and specialized controls) that will help the engineer optimize cost-effective operations and (2) information that will help the engineer stay out of trouble with legislators, particularly environmental legislators (regardless of whether the legislation is current or impending). Fines levied for ignoring emissions or pollutant statutes may not be high enough to be a deterrent in themselves when weighed against a process plant's gross production revenue. They can, however, whittle away at profits while adding to overall costs per operating hour. Frequently, though, environmental equipment can actually result in machinery's attaining longer times between overhauls. Also, the loss of goodwill —that priceless commodity on annual reports—is immense if environmental standards are not followed.

In this competitive age, plants do vie for national, state, or provincial quality control awards. Clever managers can turn those into longer customer lists. Attaining these awards is not something that many accountants, lawyers, and MBAs, who run major corporations but may have little or no technical exposure, can pull off without their engineers. It is the engineers who are likely to be the key figures in putting together the action framework for what will buy their firm new or continued goodwill. Environmental accounting plans, holistic management of resources and waste products, environmental policy, waste and toxic management—they mean pretty much the same thing and they are not a feature at all in many other process engineers' reference books.

It is painfully evident that the emphasis given to waste and toxic management varies globally. It reaches a high in Sweden and Norway, England is fast developing an aggressive proactivity in this vein, and Canada has excellent technology, which may or may not get enforced to the appropriate extent depending on the political balance of power at any time. The United States has some large loopholes that are surprising for a country so advanced; shared emissions legislation is one. And yet, it's in the area of waste and toxic management that companies receive the most vocal and widespread media coverage (and loss of business) when exposed. Some of the world's youth appear to have a sense of resources running low and therefore a need to conserve them. In these days of increasing international joint ventures, the gaps between all these preferences is fast diminishing and the stable point for the resultant system can tend to reflect the highest standards among the partners.

One could argue that subjects that infringe on environmental and waste management turf belong in another handbook and with another kind of engineer. That is not entirely true though this is becoming a specialist field. The reasons for this statement are rooted in profit margins. If environmental considerations and waste products can be integrated into production in a way that what might have been a hazard or waste now contributes to revenue, this is obviously preferable to that hazard or waste being isolated with its own disposal/neutralization system that does not contribute revenue. Some examples are biomass waste in pulp and paper production, formerly disposed of, that can be converted to gaseous fuel for a turbine (*see* Pulp and Paper) and chemical by-products in complex downstream petrochemical plastics production, otherwise waste, that can now also be used as turbine fuel. The controls and system modifications that assist incorporation of these profitable adaptions into process plants are given some space.

I have also included some basic information on specific controls and monitoring systems. They are a fact of life on a process engineer's turf; the ones I have highlighted have a proven track record for adding profit margins to processes by minimizing downtime or fluctuations.

Similarly, a process engineer may have to make decisions related to turbomachinery performance or capacity that are affected by metallurgical processes. Included is some information on common critical alloys used in today's plants.

This book contains information on the major components and basic systems, including instrumentation and controls, and some optimization techniques that I wish I had had when I landed, albeit happily, in my first major plant. It also contains examples, drawn from knowledgeable sources, of action plans that have kept various process companies in good standing and high esteem with their public and governments worldwide. Selected extracts of the technology that are the bases of these policies are also included. These examples and technology extracts are frequently missing from engineering handbooks; I would be doing users of this handbook a disservice to leave out this information.

Increasingly process plants are becoming small power producers. Governments are now beginning to offer incentives to small power producers. The Thai government, which buys the excess power from Esso's Sriracha refinery, is just one such example. The Alberta, Canada, government buys excess power from Syncrude Canada Limited, which produces 170,000 barrels of crude oil a day. The British power network buys excess power from Elf Acquitaine's Flotta terminal, which collects North Sea petroleum products.

In other words, this book aims to provide a process engineer with:

- Knowledge of the basics the process engineer will meet up with
- Enough knowledge to help the process engineer optimize operation safety, efficiency, and profit margins
- Information about environmental systems and avoiding trouble with the law
- Tools to integrate the plant's operation with other services, such as power production and waste management, to further optimize profits and minimize losses due to interruptions in services provided by external companies

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